MOVING LEMMA: REPRESENTATIVES MEETING PRESCRIBED SUBVARIETIES PROPERLY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original section 0B1N: sources/stacks-chow/tags.txt, source lines 13901–13901. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B1P: sources/stacks-chow/tags.txt, source lines 13902–13902. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B1Q: sources/stacks-chow/tags.txt, source lines 13903–13903. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0H8A: sources/stacks-chow/tags.txt, source lines 21485–21485. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B1R: sources/stacks-chow/tags.txt, source lines 13904–13904. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B1J: sources/stacks-chow/tags.txt, source lines 13897–13897. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GW: sources/stacks-chow/tags.txt, source lines 3243–3243. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $X \subset \mathbf{P}^N$ be a nonsingular closed subvariety. Let $n = \dim(X)$ and $0 \leq d, d' < n$. Let $Z \subset X$ be a closed subvariety of dimension d and $T_i \subset X$, $i \in I$ be a finite collection of closed subvarieties of dimension d'. Then there exists a subvariety $C \subset \mathbf{P}^N$ such that C intersects X properly and such that*

$$C \cdot X = Z + \sum\nolimits_{j \in J} m_j Z_j$$

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where $Z_j \subset X$ are irreducible of dimension d, distinct from Z, and

$$\dim(Z_j \cap T_i) \leq \dim(Z \cap T_i)$$

with strict inequality if Z does not intersect T_i properly in X.

Proof. Write $\mathbf{P}^N = \mathbf{P}(V_N)$ so $\dim(V_N) = N+1$. We are going to choose a sequence of projections from points

$$\begin{aligned} & r_N : \mathbf{P}(V_N) \setminus \{p_N\} \to \mathbf{P}(V_{N-1}), \\ & r_{N-1} : \mathbf{P}(V_{N-1}) \setminus \{p_{N-1}\} \to \mathbf{P}(V_{N-2}), \\ & \ldots, \\ & r_{n+1} : \mathbf{P}(V_{n+1}) \setminus \{p_{n+1}\} \to \mathbf{P}(V_n) \end{aligned}$$

as in Section 0B1N. At each step we will choose $p_N, p_{N-1}, \ldots, p_{n+1}$ in a suitable Zariski open set. Pick a closed point $x \in Z \subset X$. For every i pick closed points $x_{it} \in T_i \cap Z$, at least one in each irreducible component of $T_i \cap Z$. Taking the composition we obtain a morphism

$$\pi = (r_{n+1} \circ \ldots \circ r_N)|_X : X \longrightarrow \mathbf{P}(V_n)$$

which has the following properties

(1) π is finite,
(2) π is étale at x and all x_{it},
(3) $\pi|_Z : Z \to \pi(Z)$ is an isomorphism over an open neighbourhood of $\pi(x_{it})$,
(4) $T_i \cap \pi^{-1}(\pi(Z)) = (T_i \cap Z) \cup E_i$ with $E_i \subset T_i$ closed and $\dim(E_i) \leq d + d' + 1 - (n+1) = d + d' - n$.

It follows in a straightforward manner from Lemmas 0B1P, 0B1Q, 0H8A, and 0B1R and induction that we can do this; observe that the last projection is from $\mathbf{P}(V_{n+1})$ and that $\dim(V_{n+1}) = n+2$ which explains the inequality in (4). In a bit more detail, let $X_j, Z_j, T_{i,j} \subset \mathbf{P}(V_j)$ be the images of X, Z, T_i for $j = N, \ldots, n$. Then $T_{i,j+1} \cap r_{j+1}^{-1}(Z_j) = (T_{i,j+1} \cap Z_{j+1}) \cup E_{i,j+1}$ where we have a dimension bound on $E_{i,j+1}$ from Lemma 0B1R. Then note that $T_i \cap \pi^{-1}(\pi(Z))$ is the union of $T_i \cap Z$ and the inverse images of the $E_{i,j}$. Since the transition morphisms between the X_j are finite we obtain the desired dimension bound.

Let $C \subset \mathbf{P}(V_N)$ be the scheme theoretic closure of $(r_{n+1} \circ \ldots \circ r_N)^{-1}(\pi(Z))$. Because π is étale at the point x of Z, we see that the closed subscheme $C \cap X = \pi^{-1}(\pi(Z))$ contains Z with multiplicity 1 (local calculation omitted). Hence by Lemma 0B1J we conclude that

$$C \cdot X = [Z] + \sum m_j [Z_j]$$

for some subvarieties $Z_j \subset X$ of dimension d. Note that

$$C \cap X = \pi^{-1}(\pi(Z))$$

set theoretically. Hence $T_i \cap Z_j \subset T_i \cap \pi^{-1}(\pi(Z)) \subset (T_i \cap Z) \cup E_i$. For any irreducible component of $T_i \cap Z_j$ contained in E_i we have the desired dimension bound. Finally, let V be an irreducible component of $T_i \cap Z_j$ which is contained in $T_i \cap Z$. To finish the proof it suffices to show that V does not contain any of the points x_{it}, because then $\dim(V) < \dim(Z \cap T_i)$. To show this it suffices to show that $x_{it} \notin Z_j$ for all i, t, j.

Set $Z' = \pi(Z)$ and $Z'' = \pi^{-1}(Z')$, scheme theoretically. By condition (3) we can find an open $U \subset \mathbf{P}(V_n)$ containing $\pi(x_{it})$ such that $\pi^{-1}(U) \cap Z \to U \cap Z'$ is an isomorphism. In particular, $Z \to Z'$ is a local isomorphism at x_{it}. On the other hand, $Z'' \to Z'$ is étale at x_{it} by condition (2). Hence the closed immersion $Z \to Z''$ is étale at x_{it} (Morphisms, Lemma 02GW). Thus $Z = Z''$ in a Zariski neighbourhood of x_{it} which proves the assertion. □